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Estimating Abundance and Survival in the Endangered Point Arena Mountain Beaver Using Noninvasive Genetic Methods

Abstract

The Point Arena mountain beaver (*Aplodontia rufa nigra*) is federally listed as an endangered subspecies that is restricted to a small geographic range in coastal Mendocino County, California. Management of this imperiled taxon requires accurate information on its demography and vital rates. We developed noninvasive survey methods, using hair snares to sample DNA and to estimate abundance and survival at two sites, Kinney Beach and Alder Creek, within Manchester State Park. We extracted DNA and genotyped 371 hair samples resulting in the identification of a total of 54 individuals during annual sampling from 2006–2009. Estimated population numbers were small, ranging from 9–18 individuals at Kinney Beach and 14–18 individuals at Alder Creek. Neither location demonstrated a trend in abundance over the 4-year sample period. There was weak support (evidence ratio 2.15) for higher apparent survival probabilities at Alder Creek (0.75) than Kinney Beach (0.59) and no support for time or site effects on recruitment. Recruitment ranged from 0.25 to 0.46 and was highest during the same interval (2007–2008) at both locations. The time series of estimates from 2006–2009 does not suggest that abundance at either study site is declining; while reassuring, concern still remains due to low total numbers at this, one of the few protected sites for this endangered subspecies.

Keywords: Point Arena mountain beaver, *Aplodontia rufa nigra*, California, abundance, demography, survival

Introduction

The Point Arena mountain beaver (*Aplodontia rufa nigra*; hereafter PAMB) (Figure 1) is one of seven recognized subspecies (Hall 1981) within the monotypic genus (Taylor 1918). It occupies a small, isolated, 85-km² geographic range occurring only in coastal Mendocino County, California (Steele and Litman 1998). The PAMB was listed as endangered under the US Endangered Species Act due to threats posed by land conversion and human disturbance combined with its highly restricted distribution (USDI Fish and Wildlife Service 1991). Yet despite its endangered status there is little known about its ecology, demographics, or population status (i.e., Camp 1918, Fitts 1996, Billig and Douglas 2007, Zielinski et al. 2010), thus making conservation and recovery efforts more challenging.



Figure 1. A Point Arena mountain beaver (*Aplodontia rufa nigra*) captured in southern Mendocino County, California.

Mountain beavers are semi-fossorial and herbivorous. They excavate burrow systems that include a network of tunnels along with chambers used specifically for denning, feeding, and storage of food, fecal pellets, and refuse (Voth 1968).

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Dens are in the home burrow area (Hubbard 1922) and are roughly circular chambers filled with dry vegetation formed into a nest (Camp 1918, Martin 1971). Mountain beavers typically occur in forest openings where the soil is moist and shrubs and herbaceous vegetation is dense (Feldhamer et al. 2003). PAMBs occur in these environments in the eastern portion of their range (Billig and Douglas 2007), but in the western portion they also commonly occur in non-forest environments of coastal shrub and herbaceous cover types (Steele and Litman 1998) where structurally complex vegetation is limited (Zielinski and Mazurek 2006, Zielinski et al. 2010).

There has been a relatively long history of surveys and management plans pertaining to PAMBs within Manchester State Park (e.g., Fitts et al. 2002) each one establishing the need for more scientific information on abundance and survival to assist in recovery. The development of new tools for conducting noninvasive genetic studies on PAMB population ecology (Pilgrim et al. 2006) has made it possible to design and implement a noninvasive field method with the potential to estimate abundance, survival, and movement patterns (e.g., Schwartz and Monfort 2008). Thus our goal was to use non-invasive genetic surveys to estimate the abundance of PAMBs over multiple time intervals to assess if the abundance was increasing, stable, or decreasing (Schwartz et al. 2007). We present the results of four years of genetic monitoring to estimate PAMB abundance, recruitment and survival rates at several locations in Manchester State Park, Mendocino County, California.

Study Area

This study was conducted in Manchester State Park (the Park), Mendocino County, California. The Park is located on an alluvial plain formed by the Garcia River where the topography is comprised of low hills formed by stabilized sand dunes and coastal terraces. The research study was conducted in the central and northern portion of the Park (Figure 2) where the vegetation is coastal scrub with limited surface water. The climate is Mediterranean maritime, with relatively cool

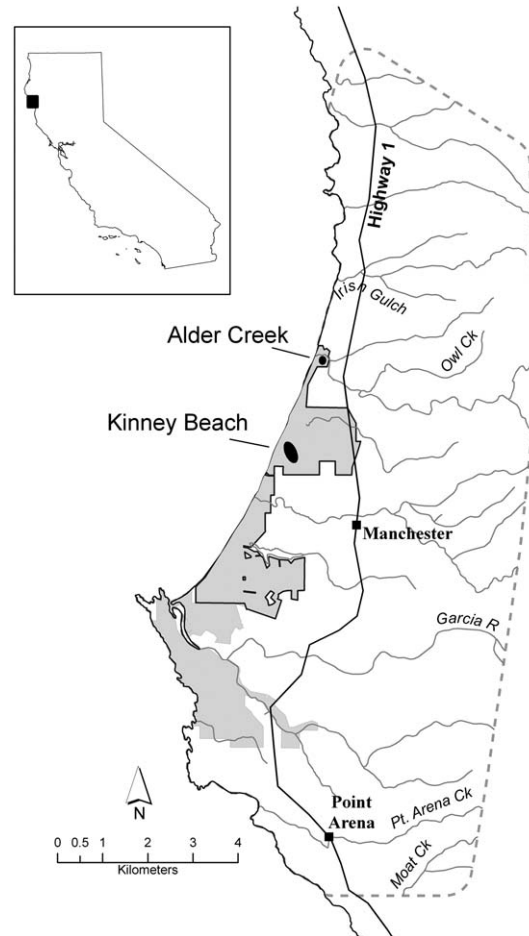


Figure 2. Map of the putative Point Arena mountain beaver (*Aplodontia rufa nigra*) range (dashed gray line) in southern Mendocino County, California. Protected public lands are shown as gray polygons, with Manchester State Park outlined in black and the study areas (Alder Creek and Kinney Beach) identified. East-west stream systems are shown as thin black lines, and the town (Point Arena) is shown as a black square.

summers (average July maximum is 18.4 °C and minimum is 9.9 °C). Winters are wet with only occasional freezing temperatures.

Methods

Sampling Design and Hair Collection

Other researchers have collected hair from burrow-dwelling species by taking advantage of the close contact between an animal and its burrow substrate

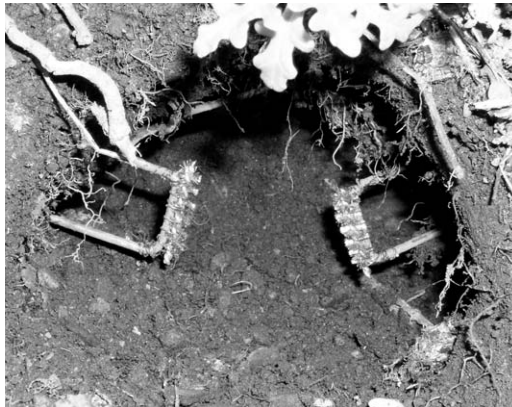


Figure 3. Hair snare set used to collect hair from Point Arena mountain beavers (*Aplodontia rufa nigra*) from 2006–2009 in Mendocino County, California. It includes two snares, each comprised of a small U-shaped landscape stake to which a gun-cleaning brush was wired. A small piece of apple placed between the snares (not pictured), was also a part of the set.

(Sloane et al. 2000, Walker et al. 2006). We improved upon this method by developing wire-brush hair snares inserted in the burrow wall (Figure 3). Preliminary tests indicated that an optimal snare design consisted of two brushes and a piece of apple for bait (Zielinski and Mazurek 2006).

We selected the Kinney Beach (Kinney) and Alder Creek (Alder) areas of the Park as study sites (Figure 2) because our previous work showed these areas to be the only Park locations with perceived densities high enough to precisely estimate abundance. Our surveys of other portions of the Park did not reveal other areas with contiguous burrow systems that were > 0.25 ha. Furthermore, genetic data suggest that these sites appear to be poorly connected to other burrow areas outside the Park (Zielinski et al. 2012).

To thoroughly sample occupied areas at each study site we established a GPS-referenced, rectilinear grid of reference stakes, spaced every 15 m and oriented along the long axis of the available habitat (as understood from earlier work; Zielinski and Mazurek, 2006). Given that the average home range size is 1369 m^2 (W. Zielinski, unpublished data) this spacing exceeds the recommended density for abundance estimation (White et al. 1982). The grid extended on each side of the long axis until we could no longer detect burrows

(Figure 4). Around each stake we searched for up to four burrow entrances that showed signs of recent activity (e.g., fresh dirt fans outside the entrance, well-trodden runways, and green vegetation pulled into the burrow). We marked burrow openings with small wire flags and placed two opposing hair snares in the interior burrow walls, usually 10–15 cm inside the entrance. Although each stake could reference up to a maximum of four burrows with snares, the actual number of snares installed near each stake depended on the number of burrow entrances available in the 7.5-m radius around it. Our goal was to estimate and monitor abundance in each area, which meant that we increased the size of our grid in a sampling season if it was evident that the occupied area had increased in size compared to the previous year. Therefore, prior to setting the snares each year, we walked the perimeter of each study site to search for burrows to determine whether the grid needed to be extended into newly occupied areas on the margin. These field reconnaissance activities led to modest increases in the number of grid cells surveyed from one year to the next (Figure 4). We never eliminated any grid cells, even if they appeared to be unoccupied; however, we did not place snares at grid points where there no longer appeared to be recent burrowing activity.

The time interval between checking the snares varied during the beginning of the first year of sampling, to reconcile the need to avoid long periods when hair samples were left in the field with the need to leave the snares in place long enough for animals to move through their burrows. In the first year only (2006), and only at Kinney, we surveyed in spring and began by checking snares 24–36 h after deployment, resulting in a poor success rate (9.7%; 12 hair samples collected from 124 burrows). We immediately extended the visit interval to three weeks and obtained a 44.0% success rate. However, we reduced the visit interval to 7–14 days for the next three visits because we did not want the DNA in our samples to degrade due to moisture. Each year thereafter, at both Kinney and Alder, we checked snares every 7–10 days (Table 1). Thus, snares at both locations were typically set out within a two-day period and then checked three times, typically between

A.

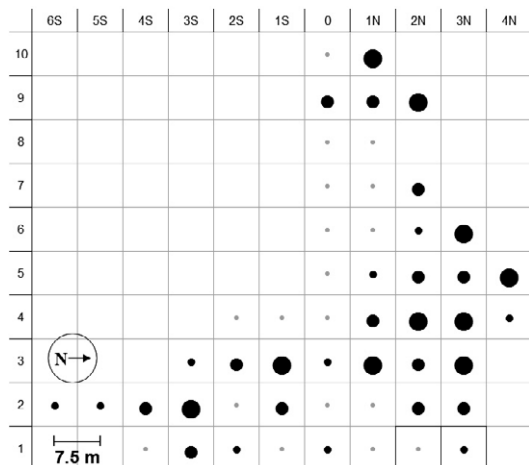
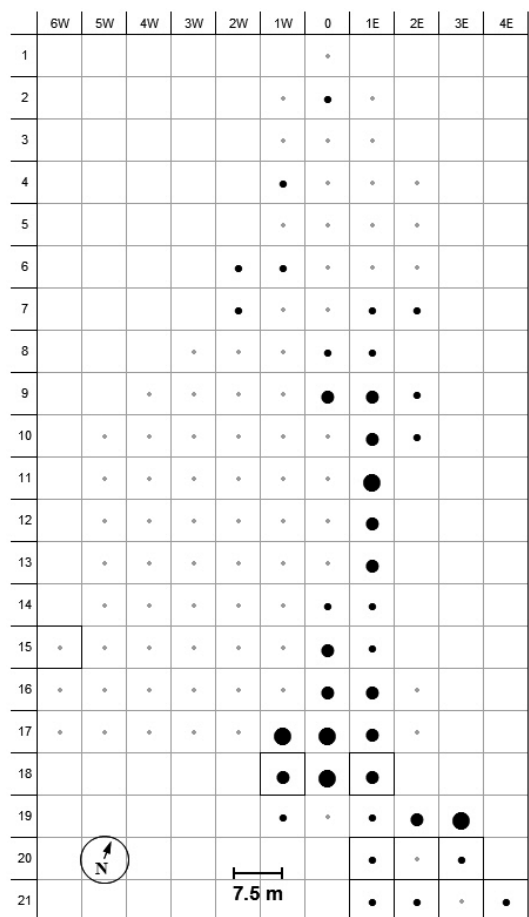


Figure 4. (left) The sampling grids and distribution of Point Arena mountain beavers (*Apodonta rufa nigra*) at Alder Creek (A) and Kinney Beach (B), in Mendocino County, California from 2006–2009. Size of circle represents the number of individuals verified at the grid point times the number of years (i.e., individual-years). Small gray circles represent no individuals captured, small black circles represent 1–2 individual-years, medium black circles represent 3–4 individual-years, and large black circles ≥ 5 or more individual-years. Black outlined grid cells were added in 2008 and 2009 due to expanded burrow activity in these areas.

B.



September and November each year (Table 1). Variations in start dates were typically due to a combination of weather and availability of field personnel. The work was conducted according to the guidelines developed by the American Society of Mammalogists for research on live animals (Gannon et al. 2007).

Hair Sample Sorting and Morphological Analysis

A hair sample represents the hairs pooled from both snares at a single burrow, collected during a single sampling interval. The number of hairs ranged from one to hundreds, but was typically > 10 . All samples were labeled in the field and placed in a silica gel indicator desiccant (Sorbead Orange or Blue to Pink, eCompressedair, Tulsa, OK) within 24h of collection, then stored in a cool, dark, airtight container. Because species other than mountain beavers use their burrows, we attempted to exclude hair from non-target species prior to laboratory analysis. Using a stereo dissecting microscope, we removed all hairs from both brushes in a single sample, discarded any extraneous vegetation or soil, and then examined the hairs for gross morphological characteristics. Using an identification key based on hair morphology (Zielinski and Mazurek 2006) we determined which hairs were most likely to be from mountain beaver and subjected only these hair samples to DNA analysis. When a hair sample included more than 10 hairs, we selected 10 hairs with the most obvious and intact follicle; when there were < 10 hairs we extracted DNA from them all. In each case the hair was pooled for a single DNA extraction. In doing so, we assumed that the hairs from each pair of snares were deposited by a single individual. This was

TABLE 1. Sampling periods and summary of Point Arena mountain beaver (*Aplodontia rufa nigra*) samples collected and analyzed at the Alder Creek and Kinney Beach sites each in Manchester Beach State Park, Mendocino County, California, 2006–2009.

Monitoring site and year	Sampling dates		Duration (days)	No. of visits	No. of burrows sampled ^a	No. of samples collected	No. of samples analyzed ^b	No. that did not amplify ^c	No. that amplified	Amplification rate ^d	Success rate ^e	No. of individuals identified ^f	Sex ratio ^g
	From	To											
Alder Creek													
2006	25 Aug	29 Sep	34	3	88	184	156	82	74	47.4%	40.2%	12	7:4:1
2007	28 Nov	19 Dec	21	3	88	124	114	85	29	25.4%	23.4%	11	5:5:1
2008	17 Sep	8 Oct	21	3	104	173	148	108	40	27.0%	27.0%	11	5:4:2
2009	30 Sep	23 Oct	23	3	103	217	205	143	62	30.2%	28.5%	14	7:6:1
Kinney Beach													
2006	10 Apr	11 May	31	3	149	210	173	130	43	24.9%	20.5%	14	5:5:4
2007	28 Aug	19 Sep	21	3	129	119	103	78	25	24.3%	21.0%	7	3:4:0
2008	16 Sep	7 Oct	21	3	127	171	142	100	42	29.6%	24.6%	12	5:6:1
2009	1 Oct	23 Oct	22	3	138	255	233	177	56	24.0%	21.9%	13	5:8:0
Totals					926	1453	1274	903	371				

^a Two hair snares were placed in each burrow entrance.
^b Number of samples collected less samples that: (1) were determined to be non-mountain beaver upon examination under a dissecting microscope, (2) contained no hair or only vegetation upon examination at the laboratory, and/or (3) contained natural or synthetic strands of material determined not to be hair.
^c Samples which, after DNA extraction, did not contain enough quality DNA to obtain individual identification and/or sex or had adequate DNA but did not produce any amplicons.
^d Number of samples with individual identification / number of samples analyzed.
^e Number of samples with individual identification / number of samples collected.
^f Total number of individuals detected per year.
^g Males:Females:Unknown.

supported by the failure to observe three alleles at any locus, which would have been an indication of a mixed sample. The variability of our sample and the large number of markers we used would have made this observation likely if snare samples included hair from multiple individuals. DNA was extracted within two months of sample collection in the field.

Genetic Methods: Individual and Sex Identification

Genomic DNA from hair samples was extracted following the protocols in Mills et al. (2000), which is a modified QIAGEN DNA extraction protocol (hair samples are incubated overnight at 55 °C horizontally in tubes on a rocker such that the entire length of the hair is bathed in the ATL buffer). All hair samples were first screened with three microsatellite loci that only amplify in mountain beavers and fail to amplify in other species (*A12*, *A104*, *C3*; Pilgrim et al. 2006). In this manner we saved time and money by not having to perform a species identification test first, and we were able to exclude mountain beaver hair samples that did not contain high quality DNA. Samples that amplified successfully were then genotyped at the remaining 10 loci for a total of 13 polymorphic loci (9 from Pilgrim et al. [2006]: *A1*, *A12*, *A104*, *A114*, *B8*, *B12*, *C3*, *C6*, *D101*, and four from Piaggio et al. [2009]: *ArE04F*, *ArA08F*, *ArH04F*, *ArG05F*). DNA was amplified with polymerase chain reaction (PCR) using a reaction volume of 10 µL containing 2.0µL DNA, 1x reaction buffer (Perkin-Elmer, Waltham, MA), 2.5 mM MgCl₂, 200µM of each dNTP, 1µM reverse primer, 1µM dye-labeled forward primer, 1.5 mg/ml bovine serum albumin (BSA), and 1U *AmpliTaQ Gold* polymerase (Applied Biosystems, Foster City, CA). The PCR profile was 94 °C/5 min, [94 °C/1 min, X/1 min, 72 °C/30s] for 45 cycles, where X is 53–60 °C depending on the specific primer. Primers had the 5' end labeled with either IRD700 or IRD800 dye for visualization on the Li-Cor sequencer. The PCR products were run in a 6.5% acrylamide gel for two hours on a Li-Cor DNA analyzer (Li-Cor Biotechnology, Lincoln, NE).

Sex determination was performed using the zinc-finger region of the X and Y-chromosomes

amplifying with general mammalian primers (García-Muro et al. 1997) and a sex specific test developed for mountain beavers (Pilgrim et al. 2012).

Genotype Error Checking

We detected genotyping errors using multiple approaches. First, each sample was amplified three times at each locus using a multi-tube PCR approach (Taberlet et al. 1996) and scored by two independent observers. Samples that failed completely, or were inconsistent between the three amplifications, were re-extracted and re-amplified. If no additional hairs remained for re-extraction we only re-amplified all loci that failed or provided inconsistencies. Samples were culled from analysis (and considered to “fail to amplify”) if after the second round of reanalysis they failed or provided inconsistent results at > four loci.

Next, we conducted the EB test and the DCH test (McKelvey and Schwartz, 2004a,b) using program DROPOUT (McKelvey and Schwartz 2005). Samples that only differed by one or two loci (the EB test) were re-checked for accuracy and re-amplified (Schwartz et al. 2006). This process was continued until the DCH test identified no additional errors. Third, every confirmed genetic result was then spatially plotted in ArcMap (ESRI, Redlands, CA) to ensure the results were in the same study area and within about 150 m (a distance that represents the diameter of most of the home ranges; Zielinski, unpublished data) of other locations with the same genotype (similar to Smith et al. 2006, Marucco et al. 2009). The few individuals that differed at less than 2 loci but were within about 150 m were reanalyzed. Last, we used microchecker (van Oosterhout et al. 2004) to catch any additional genotyping errors.

Estimating Abundance and Survival

Abundance, survival, and recruitment were estimated using a Pradel robust design model with Huggins closed captures in program MARK (White and Burnham 1999). We considered capture occurred when an individual was identified from DNA in snared hair. Robust models assume closure between secondary capture periods but allow individuals to enter and leave a population

(i.e., open populations), between primary capture periods. In our case, the primary capture periods were years 2006, 2007, 2008, and 2009, and secondary capture periods were the three consecutive weekly capture occasions that occurred during the fall of each year. By using closed population estimators for abundance and open estimators for apparent survival (and a combination of these for recruitment), capture heterogeneity can be examined and, if necessary, included in the analysis. Each hair sample that was reliably genotyped was considered a capture or recapture, and we created a capture history for each individual for each primary and secondary capture period. To obtain unbiased estimates of recruitment, it generally is necessary to maintain stable study area boundaries over time (Nichols et al. 2000). However, in our case the aggregation of burrows that we sampled was isolated and the increase in the size of the study area reflected observed changes in burrow occupancy around the edges of the original study area.

Models were developed first by examining various approaches to modeling capture probability and then considering plausible models of survival and recruitment. We considered five approaches to modeling capture probabilities: variation across years (i.e., primary capture periods [t]), variation by site (i.e., Alder or Kinney), constant capture probability ($\cdot$), two mixture heterogeneity (2), and two mixture heterogeneity by site ($2 \times \text{site}$). Two mixture heterogeneity refers to the possibility that there are two classes of individuals within the sampled group of animals, each with distinct capture probabilities. The two mixture heterogeneity model performed better than the other models of capture probability; the only model that performed nearly as well was the time varying capture probability model. Most studies that use noninvasive methods to identify individuals have found heterogeneity in capture probabilities (Mills et al. 2000, Kendall et al. 2009). As there are difficulties associated with combining estimates from heterogeneity models with other model types, we only considered models with heterogeneity in capture probabilities when modeling survival and recruitment. Because of the limited sample size, we only considered models with $\leq$

10 parameters. In our initial analyses, models of apparent survival and recruitment included all combinations of time, site, and constant effects for both parameters. However, survival estimates were problematical (close to 1.0 with extremely large standard errors for the last interval) for all models that included time variation in survival and, therefore, those models were dropped from the model set. This resulted in a total of six models that were considered.

Models were compared using Akaike's Information Criteria corrected for small sample size (AIC_c) (Burnham and Anderson 2002). Model averaging was used to account for model selection. Confidence intervals on N were computed using unconditional estimates of variance and adjusted for M_{t+1} , the number of distinct animals captured. This constrains the lower confidence interval to be no lower than M_{t+1} (Williams et al. 2001:304).

Density estimates were calculated for the purpose of comparing densities in our study area with those reported for mountain beavers elsewhere. We estimated density by dividing the area of each site by the number of individuals estimated to occur there each fall. The size of each area was calculated by adding a 7.5-m buffer to the grid points at the perimeter of the grid: a distance equivalent to half the distance between grid points (Dice 1938). Variance estimates for density were computed using the delta method: $\text{var}(\text{Density}) = (1/\text{Area})^2(\text{var } N)$.

Results

Sampling

Sampling from 2006–2009 resulted in the collection of 1453 total samples, 698 and 755 from Alder and Kinney, respectively (Table 1). A total of 371 of these samples (25.5%) provided useful PAMB DNA for genotyping, resulting in the identification of 24 and 30 unique individuals in Alder and Kinney, respectively. We used an average of 5.52 (SD = 3.49) hairs per extraction. Samples that successfully amplified used an average of 6.6 (SD = 3.2) hairs, while those that failed used 5.0 (SD = 3.5) hairs. The low success rate could be partially explained by the fact that some samples collected were not PAMB hair (i.e., hair from the

brush rabbit (*Sylvilagus bachmanii*) minute pieces of vegetation or clothing fibers). However, a total of 32.5% and 25.7% of known PAMB samples from Alder and Kinney respectively failed to amplify sufficiently or had inadequate DNA to be genotyped (see Table 1 for rates per year).

Descriptive Population Genetic Analyses

Our microsatellite analysis produced 25 alleles at Alder and 35 alleles at the Kinney site (range 1–3 per locus). The sample of animals from Alder had an average observed and expected heterozygosity of 0.323 (SE = 0.073) and 0.291 (SE = 0.060), respectively. Animals from Kinney had an average observed and expected heterozygosity of 0.490 (SE = 0.034), and 0.500 (SE = 0.036), respectively. F_{IS} was -0.095 at Alder and 0.004 at Kinney. The negative F_{IS} is consistent with general trends observed across mammals due to social structure (Storz 1999). These 13 microsatellites provided a probability of identity of 3.44×10^{-8} and probability of identity for siblings of 4.15×10^{-4} (Paetkau and Strobeck 1994), when we include all samples in our database, independent of the effects of genetic structure. Considering only those samples from each location the probability of identity for Alder was 3.84×10^{-4} and the probability of identity for Kinney was 3.21×10^{-7} . We found no evidence for gametic disequilibrium at the Alder site and the

non-random association between alleles at locus *A1* and *C3*, *A12* and *B8*, and *A104* and *D101* at the Kinney site after Bonferroni corrections. Additional population genetic summary statistics, particularly in regard to substructure, can be found in Zielinski et al. (2012).

There was no evidence of preferential amplification of small alleles at any locus or evidence for large allele dropout. Sex ratios were relatively even at both Kinney and Alder (Table 1) and no individual was detected at both sites.

Modeling Abundance, Survival And Recruitment

No single model received strong support so models were averaged to generate parameter estimates (Table 2). Survival varied by site, but not time, in the top three models with parameter estimates of 0.75 for Kinney and 0.59 for Alder (Table 3). However, due to high model selection uncertainty, the support for a difference in survival between the sites was weak (i.e., the evidence ratio between the probability of the top model that included a site effect on survival (0.287) and the equivalent model without a site effect (0.133) was 2.51; Table 2). Recruitment varied by time in the top model but the AIC_c value was almost identical for the second model with constant recruitment. Recruitment varied by site in the third model (Table 2).

TABLE 2. Comparison of models of mark-recapture data from Point Arena mountain beavers (*Aplodontia rufa nigra*) collected using non-invasive methods at the Kinney Beach and Alder Creek sites in Manchester State Park, Mendocino County, California, 2006–2009. Model fit evaluated using Akaike's Information Criterion, corrected for small samples (AIC_c). Parameters included apparent survival (Φ), recruitment (f), initial capture probability (p), and mixture (π).

Model ¹	AIC_c ²	ΔAIC_c ³	w_i ⁴	K ⁵	Dev. ⁶
$\Phi(\text{site}) f(t) \pi(2) p(.)$	605.224	0	0.287	8	588.442
$\Phi(\text{site}) f(.) \pi(2) p(.)$	605.279	0.055	0.279	6	592.827
$\Phi(\text{site}) f(\text{site}) \pi(2) p(.)$	606.742	1.517	0.134	7	592.136
$\Phi(.) f(t) \pi(2) p(.)$	606.760	1.536	0.133	7	592.154
$\Phi(.) f(.) \pi(2) p(.)$	606.920	1.696	0.123	5	596.600
$\Phi(.) f(\text{site}) \pi(2) p(.)$	608.999	3.775	0.039	6	596.548

¹ Models names follow Lebreton et al. (1992), f varied by site (site), time (t), or was constant (.). Survival estimates were problematical for models that included $\pi(t)$ so they were dropped from the model set. A two mixture heterogeneity model was used for capture probabilities.

² Akaike's Information Criteria corrected for small sample size.

³ Difference in AIC_c from top model.

⁴ Model weight.

⁵ Number of parameters in model.

⁶ Model deviance

TABLE 3. Model-averaged parameter estimates of survival probabilities (Φ), per capita recruitment (f), capture probabilities (p), unconditional standard errors (SE), and lower (LCL) and upper (UCL) 95% confidence intervals of Point Arena mountain beavers (*Aplodontia rufa nigra*), Manchester State Park, Mendocino County, California (2006–2009). All models were run using the Robust Design Pradel Recruitment Huggins' Closed Captures with Heterogeneity data type in program MARK.

Parameter ¹	Estimate	SE	LCL	UCL
Φ (Alder Creek)	0.749	0.089	0.542	0.883
Φ (Kinney Beach)	0.594	0.096	0.402	0.762
f (Alder Creek 2007)	0.232	0.138	0.062	0.580
f (Alder Creek 2008)	0.443	0.229	0.114	0.831
f (Alder Creek 2009)	0.255	0.116	0.094	0.532
f (Kinney Beach 2007)	0.249	0.149	0.065	0.612
f (Kinney Beach 2008)	0.461	0.220	0.131	0.829
f (Kinney Beach 2009)	0.273	0.125	0.098	0.564
Π	0.428	0.124	0.216	0.669
p_1	0.223	0.126	0.065	0.543
p_2	0.782	0.069	0.619	0.888

¹ The same approach was used when estimating p in all models (heterogeneity in capture probabilities with two mixtures, no variation by time or location), hence there was only one estimate for p and Π . Φ is survival from the year in the parentheses to the next year; f is recruitment from the previous year to the year listed in the parentheses.

Model-averaged parameter estimates of survival were higher at Alder than Kinney, for all years, but confidence intervals overlapped. Recruitment was highest at both sites during the same interval (2007–2008), but the confidence intervals were large and broadly overlapped (Table 3). The mixture parameter and associated capture probabilities indicated that most of the individuals (57.2%; 1- Π ; Table 3) had a high capture probability (0.78); a sizeable minority (42.8%) had a substantially lower capture probability (0.22) (Table 3).

Point estimates of abundances were low with modest year-to-year variation (14–18 individuals at Alder and 9–18 individuals at Kinney) (Figure 5). The confidence intervals around each estimate were such that neither site exhibited a significant trend in estimated abundance over the 4-year period (Figure 5).

Densities (animals/ha) were substantially higher at Alder than at Kinney, with a mean (SE) of 14.18 (0.75) compared to 5.90 (0.68), respectively (Table 4). The distribution of animals at each site varied within each study area (Figure 4). Animals were fairly evenly distributed across the Alder site (Figure 4A). Most of the occupancy at Kinney was

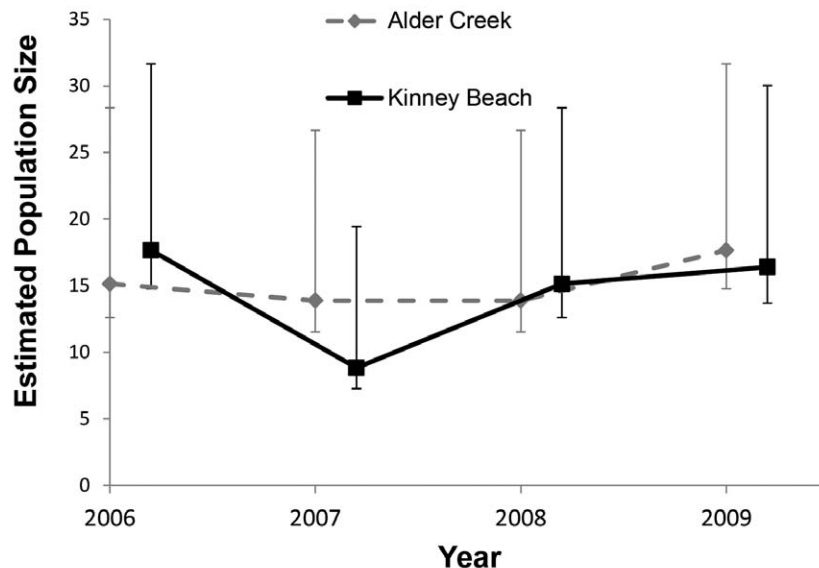


Figure 5. Model averaged estimates of Point Arena mountain beaver (*Aplodontia rufa nigra*) at Alder Creek and Kinney Beach study sites, in Manchester State Park, Mendocino County, California from 2006–2009. Error bars are unconditional 95% confidence intervals adjusted for minimum number alive, following Williams et al. (2001).

TABLE 4. Point Arena mountain beaver (*Aplodontia rufa nigra*) density estimates, based on point estimates for *N* (abundance) for Alder and Kinney sites, Manchester State Park, Mendocino County, California, 2006–2009.

	Area (ha)	<i>N</i>	Density (<i>N</i> /ha)
Alder Creek			
2006	1.05	15.2	14.48
2007	1.05	14.0	13.33
2008	1.10	14.0	12.73
2009	1.10	17.8	16.18
Mean		15.2	14.18
SE		0.77	0.75
Kinney Beach			
2006	2.37	17.8	7.51
2007	2.37	8.9	3.76
2008	2.57	15.2	5.91
2009	2.57	16.5	6.42
Mean		14.6	5.90
SE		1.71	0.68
Grand Mean			10.04
SE			2.33

centered on a north-south oriented area (Figure 4B), but during the course of our sampling over the years the northern portion of the grid became vacant. There appeared to be increased use of the southeast portion of the Kinney site near where a campground was recently removed, which was the reason that we expanded the grid slightly in the last two years.

Discussion

Although we sampled the largest areas occupied by mountain beavers in the Park, our best estimates were fewer than 20 individuals at each site during each of the four years of this study. The low abundance estimates are a concern because the Kinney and Alder sites occupy a small proportion of the Park area (~4 ha of 600 ha; 0.67%) and because the Park is one of only a few protected areas within the range of the PAMB. We are also concerned because the estimate at Kinney may have decreased to a low of nine individuals in 2007. The Kinney site was included as a control in a previous five-year study to monitor burrow distribution on a nearby (<500m) site. From

1992–1997, approximately 10 years prior to our work, the authors of that study believed that Kinney enjoyed an increase in burrow activity and presumably abundance (Northen and Fitts 1998). We do not know how many individuals occurred at the Kinney site during this earlier period, but it appears that the site has been occupied since at least 1989 (Steele and Litman 1998). This low abundance estimate is consistent with an extremely small effective population size estimate (from a linkage disequilibrium estimator) of 7.7 for Kinney Beach (Zielinski et al. 2012). Effective population sizes of this magnitude are of concern as genetic drift will be strong limiting the ability of the population to adapt to future change (Hare et al. 2011).

Several metrics point to a more secure status for the PAMBs at Alder than at Kinney. Despite a similar range in estimates of abundance, PAMBs were three times as dense at Alder as at Kinney. Effective population size estimates at Alder were also greater, although not by much (11.7; Zielinski et al. 2012). The Alder site hosts a rich native forb community that provides a diversity of foods and den materials and is comprised of loamy soils that may be more conducive to burrowing than the sandy soils at Kinney. Although there is no obvious water throughout most of the site (like Kinney), the northern margin of the Alder site includes a small wetland. This collection of favorable conditions may be responsible for the fact that the Alder site has likely been occupied for at least 100 years (Camp 1918, USDI Fish and Wildlife Service 2009).

The limited sample size precluded identifying strong effects on survival and recruitment. There was weak support for higher survival at Alder and no evidence for effects of either time or site on recruitment. Although the model averaged estimates suggested a substantial increase in recruitment during the 2007–2008 interval at both sites the standard errors of all the estimates of recruitment were large and the 95% confidence intervals were broadly overlapping. The estimates suggest that recruitment is sufficient to replace losses due to mortality over the short term but the low abundances at both sites makes them vulnerable to stochastic effects on demographic parameters.

The densities reported here, for both the Kinney (mean = 5.90/ha) and Alder (mean = 14.18/ha) sites, were much higher than previously reported for mountain beavers almost everywhere else. Densities vary within study area and year, but have ranged from 0.49/ha (Arjo et al. 2007), to 3.4/ha (Neal and Borrecco 1981), to 7.6/ha (Lovejoy and Black 1979) to 14.0/ha (Morris et al. 1995). Voth (1968) removed 21 animals in a 2 ha area (i.e., 10.5 individuals/ha) but estimated that he had failed to capture as many as 63 additional animals (i.e., a density of 42 individuals/ha). Excluding the latter estimate, which is suspect because removal of individuals may have led to immigration, our densities at Alder are the highest reported. This is particularly surprising given that we estimated abundance in the fall when densities are typically much lower than they are in the spring and summer (Voth 1968). Similarly, our annual survival rates, particularly at Alder (mean = 0.75), exceeded previous reports (i.e., 0.64, Lovejoy and Black [1979]; 0.36–0.57, Arjo et al. [2007]). Thus, the values for density and survival in our study areas suggest a more healthy status for the PAMBs in each area than do the estimates of abundance.

Our sampling protocol achieved relatively high capture probabilities, but we experienced relatively low success at amplifying PAMB DNA from hair samples. Some of this occurred because some hairs were from species other than PAMB and did not amplify using our PAMB-specific primers. Nonetheless, there is room for improvement in collecting better quality samples from the field and in extracting useful DNA (e.g., improving number of hairs snared, checking snares more frequently, extracting DNA from more than 10 hairs per snare sample). The low amplification rates probably did not affect our abundance and survival estimates because our method placed multiple snares within every likely home range, leading to relatively high capture probabilities each year. The majority of animals were captured with a probability of 0.78 for each capture occasion; the remainder were captured with a probability of 0.22. Thus, the probability of capturing an individual during an entire 3-week primary capture session ranged from 0.99 ($1 - [1 - 0.78]^3 = 0.99$) to 0.52 ($1 - [1 - 0.22]^3 = 0.52$). Finally, the role of

relatedness among individuals should be explored further, because a companion study (Zielinski et al. 2012) revealed relatively high levels of inbreeding that may be due to the structure typically found in social mammals (Storz 1999).

Our modeling results also pointed to some issues that may require additional attention in the future. For example, when we compared models of capture probability, the most parsimonious model included capture heterogeneity, which is not uncommon in mark-recapture estimates that employ non-invasive methods (Kendall et al. 2009; Marucco et al. 2010). We revealed two groups of animals in respect to capture probabilities: those with high probabilities (0.78) and those with low probabilities (0.22). This may be due to unequal distribution of hair snares within home ranges, with one group that had many snares and another group that had fewer snares within their ranges. Identifying approaches that minimize heterogeneity in capture probabilities would increase precision of our mark-recapture estimates.

Despite the small numbers at each location, the time series of estimates from 2006–2009 does not suggest that abundance at either site is declining. Although this is reassuring, when numbers are this low any significant annual variation can increase vulnerability to genetic and demographic stochasticity, increasing the risk of extirpation. Annual genetic monitoring, at least for some period, would help determine the proportion of recruits that are born at each location, versus those that arrive as immigrants.

Our information from the Kinney and Alder sites is necessary, but far from sufficient to determine the status of the subspecies. For example, we have no knowledge of the proportion of the geographic range that is occupied or the density of burrow areas where PAMBs occur outside the Park. New information on range-wide distribution should be collected. It would provide perspective when it comes to deciding what actions should be taken should abundance decline in the Park. If, for example, PAMBs are well distributed across their range, and the animals in the Park represent only a minority of occupied locations, then perhaps surveillance of the Park sites can be less frequent

and less comprehensive. However, if Alder and Kinney represent one of only a few significant occupied sites within the entire range, then their status is one that may call for close, perhaps annual, monitoring.

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